

SPECIFIC AIMS

Autism spectrum disorder (ASD) is a lifelong, neurodevelopmental disorder (NDD) that affects ~2.7% of the population and significantly impacts infant and child development (social, language, cognitive, adaptive skills), youth and adult mental health (anxiety, depression, suicidality), and overall well-being, thus representing a pressing public health challenge¹⁻⁶. Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the individual, enhancing the quality of life for the family, and reducing a lifelong financial cost^{7,8}. Unfortunately, with ASD diagnoses and therefore intervention onset often occurring after age 4-6 years^{9,10}, children are missing out on critical periods of very early intervention that could capitalize on peak neuroplasticity within the first 2 years of life¹¹⁻¹³. Accordingly, significant resources have been dedicated to the search for very early biomarkers that will identify ASD within the first year of life, and to the development of very early, prediagnostic interventions. These research priorities have the potential to improve long-term outcomes for autistic individuals and their families, but carry complex ethical considerations, especially during the exceptionally vulnerable neonatal and postpartum period in the first year of life. Ethical conduct of research¹⁴ requires participant-driven processes to determine responsible and respectful protocols and procedures that are aligned with the values of the community. **However, our field currently lacks the essential foundation of evidence-based, participant-driven clinical research guidelines that are tailored to the needs and priorities of parents and the autism community– particularly for infants within their first year of life**¹⁵⁻¹⁸. This research gap is a barrier to scientific progress – underscored by **surprisingly low rates of enrollment into early infant clinical trials** for infants at-risk for ASD and other NDDs, with lower enrollment for lower income families, resulting in biased, non-representative samples¹⁹⁻²³. These findings necessitate a critical examination of parental attitudes and health disparities in the context of ASD infant identification and intervention¹⁹⁻²⁴. Accordingly, this proposal aims to use participant voices to **develop a novel evidence base of acceptable, participant-driven methods for the design, conduct, and translation of ASD infant biomarker and pre-diagnostic infant intervention research**. Significantly, findings will inform best practice guidelines and accelerate the translation of infant ASD identification and intervention into clinical practice.

The proposed R21 supports the scientific goals of the PI's ongoing research program, which aims to examine infant ASD biomarkers and develop novel conceptual frameworks for very early infant intervention. The proposed specific aims represent a critical step in infant autism research, addressing key priorities by evaluating the value of early identification and intervention to the community, and providing novel evidence for equitable research and clinical guidelines that are aligned with diverse family values, needs, attitudes, and preferences.

Aim 1. Determine parent acceptability (perceptions, attitudes, preferences) of ASD early identification research and practice for infants < 12 months. Analyses will include effects of: infant age at screening (e.g., 1 vs. 6 months), screening method (e.g., behavioral observation vs. eye tracking vs. brain imaging), certainty of results (e.g., sensitivity/specificity), and sociocultural context (e.g., race/ethnicity).

Aim 2. Evaluate parent acceptability (perceptions, attitudes, preferences) of early-phase development studies and real-world uptake of prediagnostic ASD interventions for infants < 12 months. Analyses will include effects of: infant age at onset of intervention, target of intervention (parenting skills/psychoeducation vs. intervention content (e.g., parent-mediated vs. clinician-implemented), intervention dosage (e.g., 1x/week vs. 1x/month), and setting/context.^{19,20}

In a mixed-methods design, qualitative data will be collected from N=40 parents, analyzed via rapid qualitative analysis²⁵, and used to inform and refine our quantitative assessment, which draws from existing literature and theoretical frameworks²⁶⁻³¹. The quantitative assessment will be administered to an expanded sample of N=100 participants that will be enriched for sociodemographic representation. This will facilitate examination of the influence of demographic factors, such as race/ethnicity and maternal education, on parent perspectives. Furthermore, participants will be stratified by two variables to capture diverse experiences and contexts. Stratification by ***lived experience*** will enable comparison between parents with lived experience with ASD (i.e., those with a child with ASD) and parents with no family history of ASD. Stratification by ***current context*** will allow for comparison between perspectives of parents with infants *currently under 15 months*, who would be eligible for real-world infant ASD biomarker and intervention studies, and parents with children *currently over 3 years*, who can offer retrospective insights informed by hindsight. This comprehensive sampling approach enhances the significance and impact of this study. Findings will inform clinical research guidelines for the ethical conduct of infant ASD identification and intervention research – a crucial step in successful dissemination and uptake of evidence-based, equitable, participant-centered ASD identification and intervention practices.

SIGNIFICANCE

Given the proliferation of evidence for early indicators of ASD in the first year of life, including candidate biomarkers, the development of very early infant interventions that can be implemented with infants as young as six months is *imminent*^{18,32}. This raises a plethora of critical issues and urgently calls for ethical and pragmatic scrutiny of the conduct of autism identification and intervention research in early infancy¹⁶. Over the past decade, ethical issues related to early biomarkers, screening, and intervention for ASD have been reiterated and expanded in dozens of editorials, theoretical pieces, and white papers.^{16,18,32–45} These complex ethical considerations and implications are particularly salient when research and service implementation would occur during an exceptionally vulnerable neonatal and postpartum period, spanning the infant's birth to first birthday. Despite this loud and clear call to action, there remains an astounding **lack of participant-driven, empirical research focused on understanding and addressing the needs and priorities of the autism community in the design and implementation of very early autism identification and prediagnostic intervention studies and services**^{15–17}, particularly those designed for parents and their infant within the first year of their life. As stated over a decade ago, "Scientists need to be proactive on matters of risk and ethics. If the history of autism is any guide, failure to do so would be irresponsible."⁴¹ This proposed R21 answers this explicit call and, significantly, **addresses a critical barrier to the ethical design and implementation of infant identification and intervention research by establishing an empirical foundation for a participant-driven translational pathway**. This project is responsive to PA-21-200 by supporting optimal means of service delivery and will ultimately, address two of NICHD's Autism Research Specific Objectives: 1) improve strategies for early biomarker screening, and 2) provide methodological guidance for testing of potential interventions. Engaging participants and stakeholders is *essential* and this study will *significantly* advance our understanding of how to ethically and successfully conduct this research and translate to practice^{15,17}.

Very Early ASD Detection and Intervention in Infancy: A Mismatch in Parental Attitudes and Behaviors. Decades of research on early detection of ASD has resulted in several candidate biomarkers that can be assayed well before the emergence of clear, observable symptoms of ASD ("presymptomatic") and before a clinical diagnosis can be made ("prediagnostic")⁴⁶. Research on *early* prediagnostic biomarkers, i.e., before the age of 12 months, is a particularly high priority^{42,47} given evidence that earlier intervention leads to improved outcomes⁴⁸. However, very few studies have evaluated parent acceptability of the actual and intended use of early biomarkers, as well as acceptability of the next phase of this research – prediagnostic infant intervention studies. And the few studies addressing this issue are mixed in their results. Some studies suggest that early autism identification and intervention research is highly aligned with the values and needs of parents of autistic children^{29,49}. In light of these shared values among researchers and parents, there is a growing body of evidence that parents are reluctant to enroll in infant intervention trials and to pursue earlier diagnoses, for example:

1. Our work showed that 44% of families **declined enrollment** into an infant intervention clinical trial for 12-month-olds with prediagnostic behavioral features of ASD.²²
2. NICHD and NHGRI's BabySeq Project report that 58% of families **declined enrollment** into a newborn genomic sequencing study⁵⁰. Similarly, 57-70% of parents **decline enrollment** into NICHD's Neonatal Research Network clinical trials^{51–53}.
3. Preliminary results from our study of diagnostic experiences of Black families show that 52.5% of parents with autistic children diagnosed between 18-48 months report they **did not want an earlier diagnosis**.

Clearly, **there is not yet a full understanding of parental perceptions, attitudes, and preferences regarding infant prediagnostic identification and intervention research**^{16,49}, which could significantly hinder scientific progress. The importance of diverse parent and stakeholder perspectives is only more critical given current social movements advocating for a paradigm shift in how we conceptualize autism—from a disability/deficit model to a neurodiversity model that embraces autistic identities and promotes acceptance⁵⁴. This has led to even more heterogeneity in stakeholder perspectives about desirability of early detection and early treatment, and even further underscores the importance and significance of the proposed work. Notably, while parents are the focus of this study, autistic and neurodivergent adults and advocates will be involved in the design and implementation of this study and autistic perspectives will be the focus of a follow-up study (see **Table 1**), particularly given data suggesting that early autism screening and intervention may not be aligned with values of autistic adults^{55,56}.

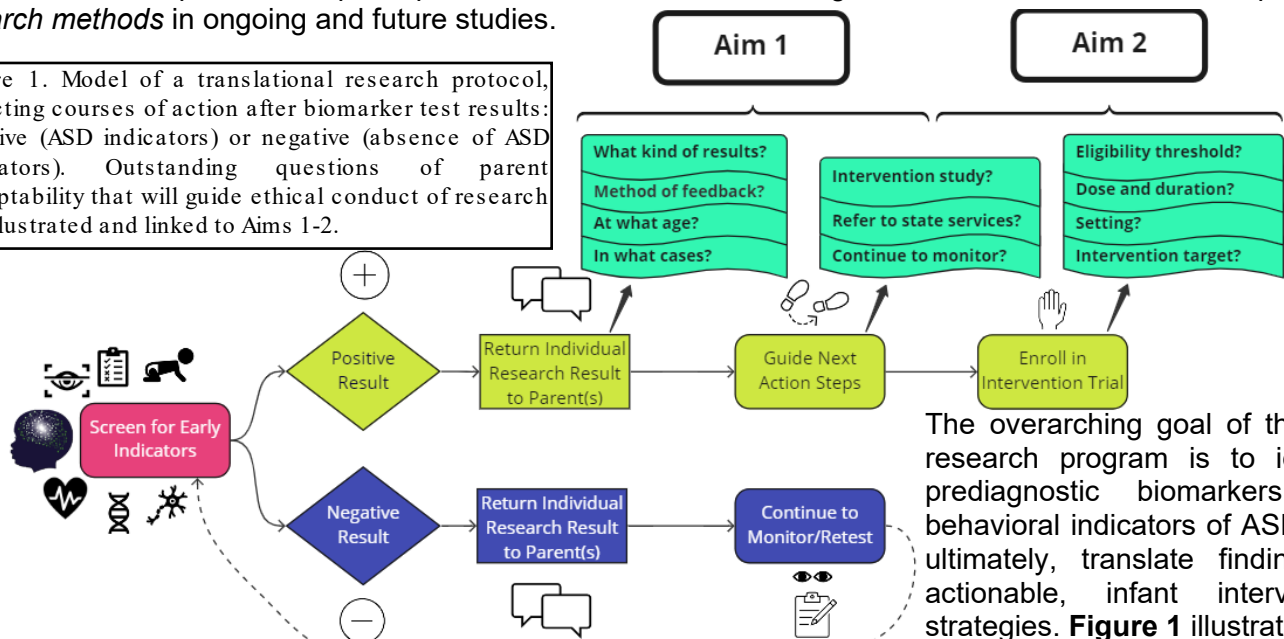
*In-depth qualitative analysis and systematic quantitative analysis of parent acceptability regarding the conduct and implementation of infant biomarker and intervention research, and subsequent translation to practice, is a **timely, crucial next step** for an autism research agenda that is ethically-informed and focused on prediagnostic identification and intervention.*^{15,16,36,42,57}

Table 1. Our Approach to Participatory Research for this Study	
Who are the participants?	Parents/caregivers are the primary decision-makers and stakeholders in the neonatal and early infant period. Thus, we focus on the <i>parents</i> as the participants in this study.
Will autistic adults/adults with ASD be included as participants?	All participants will be parents stratified by <i>lived experience</i> and <i>current context</i> (see Table 2). Parents will be asked about their own potential ASD diagnoses and autistic parents <u>will not be excluded</u> . On the contrary, autistic parent perspectives are highly valued and parent diagnosis will be considered in all analyses.
How are autistic adults and the lived experience of being autistic/having autism valued and involved in the design and implementation of this study?	Inclusion of autistic adults/adults with autism is of high value to our study team and a top priority in autism participatory research. Autistic adults/adults with autism have been full-time staff in the PI's lab for over four years and will be heavily involved in this study. Data from this study will be used to develop a follow-up study autistic adult acceptability of infant ASD identification and intervention, with the aim to include adults who can communicate their perspective <u>and</u> who represent a range of support needs.

Leveraging Participant Voices to Determine Ethical Conduct of Infant Autism Research. Established benchmarks for the ethical conduct of research¹⁴ require participant-driven processes, particularly to determine the social value of the research, the risk-benefit ratio to participants, and clear informed consent procedures. Yet, participant-driven methods have historically been ignored in autism research⁵⁸. The development of scientific protocols and procedures with human subjects without the input of the subjects themselves may have critical consequences, such as mistrust, lack of community-level impact, and *impeding scientific progress*¹⁷. The limited research thus far already highlights discrepancies between participant preferences and researcher assumptions and expectations. For example, while most parents (85%) prefer researchers provide full disclosure of all research assessments results²⁹, recent work suggests disclosure of individual research results should only be provided when findings are clinically significant and actionable⁴⁰. Similarly, the discrepancy between researcher-planned enrollment and actual enrollment into infant intervention trials²² suggests that ethical

considerations have not yet been addressed. **The current study aims to significantly improve scientific knowledge in the ethical conduct of autism in infancy research by using participant-driven methods to determine parent perspectives and preferences in the conduct of autism identification and prediagnostic intervention studies, with the goal of improving ethical conduct of research, strengthening researcher-participant partnerships and trust, improving alignment of researcher-stakeholder values, and ensuring that research and results are meaningful and beneficial to the autism community.** The goal of this study is not to develop, test, or validate specific biomarkers or treatment methods, but to develop and disseminate a much-needed, comprehensive, participant-informed evidence base to guide research *directions* and *responsible research methods* in ongoing and future studies.

Figure 1. Model of a translational research protocol, depicting courses of action after biomarker test results: positive (ASD indicators) or negative (absence of ASD indicators). Outstanding questions of parent acceptability that will guide ethical conduct of research are illustrated and linked to Aims 1-2.



The overarching goal of the PI's research program is to identify prediagnostic biomarkers and behavioral indicators of ASD and, ultimately, translate findings to actionable, infant intervention strategies. **Figure 1** illustrates a translational pipeline model for early autism research and notes questions

and research gaps to be addressed in Aims 1-2. The prior research and our preliminary data provide support for the overall study premise, research methods, and analytic plan, to determine parent acceptability that can be used to improve ethical conduct of infant biomarker and intervention studies and increase the likelihood for subsequent successful clinical translation, dissemination, and implementation.

Racial/Ethnic Representation. Participants in our ongoing infant autism study samples are significantly more representative than most studies of this kind (e.g., ~45% racial/ethnic minorities in our studies^{59,60} compared to ~15% in recent, similarly designed studies^{49,61}). This racial/ethnic diversity is critical for meaningful comparisons of both qualitative and quantitative data across sociodemographic strata, and will result in a deeper, more nuanced understanding of parent acceptability. More importantly, this approach will serve to **directly address underrepresentation of racial/ethnic minorities** in research as results will serve as a guide to inform the design of future studies that can be tailored to the unique needs of diverse communities in the US.

INNOVATION

Participant-Driven Research Methods. Participant-informed and participant-driven research is a high priority in the study of clinical populations, yet early autism researchers are falling behind in the development of participant-driven ethical research protocols⁵⁸. That is, participant-driven research methods are far from innovative in the field of public health more broadly, but existing literature points to this as a novel approach in the field of infant autism research that has the potential to shift existing paradigms of prediagnostic intervention for ASD. For example, our parent partner and neurodiversity advocate (see Letters of Support) firmly supports research and clinical practice in autism early identification and intervention and expresses enthusiasm for biomarker testing at 12 months of age, but is hesitant and resistant to testing at 6 months of age. If this sentiment is largely shared amongst stakeholders, this finding would challenge the current “earlier the better” approach and would warrant additional research that may shift current research paradigms and approaches.

Inclusion of Parents of Neurotypical and Autistic Children. To our knowledge, the limited studies evaluating parent acceptability of infant autism research have focused solely on parents/caregivers with children with ASD.^{22,29,62} Yet, infant identification and intervention research should ultimately have a population-based focus, including infants with *and without* a known family history of ASD. For findings to be generalizable, gathering perspectives of parents who are naïve to ASD is paramount. An innovative extension of prior work is a participant sample that represents parents of neurotypical children (i.e., children without ASD; “NT parents”) as well as parents with at least one child with ASD (“ASD parents”). Moreover, NT and ASD parents have distinct lived experiences. ASD parents are intimately familiar with the challenges and benefits of raising a child with ASD and most are knowledgeable about early intervention systems and the implications therein for their children and their family. NT parents are largely naïve to the lived experience of having a child with ASD but, in fact, represent the majority of parents who would receive biomarker testing and early intervention in future clinical settings. Indeed, **most ASD parents are “NT parents” until their first child is diagnosed with ASD**. Thus, these experiences offer incredibly valuable perspectives that could lead to novel insights about the conduct of early autism research. Inclusion and comparison of *both* NT and ASD perspectives is an innovative feature of this study that will greatly enhancing generalizability and impact of findings.

APPROACH

PRELIMINARY DATA

The scientific premise for the proposed research is based on our preliminary data, existing empirical work, and a significant body of research that supports the need for participant perspectives on priorities, design, and implementation of early autism research, which will inform guidelines for respectful and responsible conduct of research and practice. Our prior work demonstrates that parents enrolled in early autism identification studies are surprisingly reluctant to enroll in a prediagnostic intervention for 12-month-old infants showing early signs of ASD – over 40% of parents declined enrollment.²² Our preliminary analyses from a study on diagnostic experiences of black families (funded by the Autism Science Foundation) suggest that over half of parents *disagreed* with wanting an earlier ASD diagnosis for their child. Preliminary, retrospective interviews with parents on our Community Advisory Board suggest a preference for biomarker testing no earlier than 12 months. Other research shows that despite enthusiasm for early identification research and significant parental concern for later-born infant siblings of autistic children,⁴⁹ parents rank “designing early interventions” as a *low priority* in early autism research.^{29,62} Research on neonatal and early infant screening and intervention for non-ASD-related conditions suggest similar reluctance to enroll. The critical gap to be filled here is an in-depth understanding of parental perceptions, attitudes, and preferences, and the factors that affect overall parent acceptability:

- 1) How prevalent is reluctance to participate in early identification and intervention studies?
- 2) What perceptions or experiences are driving parent attitudes about participation?
- 3) What real or hypothetical factors affect attitudes about participation?
- 4) What are parent preferences in the design and conduct of these studies (e.g., Fig. 1)?

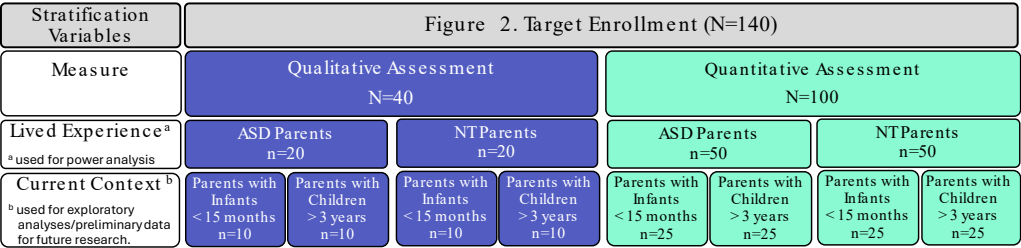
The proposed study leverages existing frameworks^{26–28} and measures parent perspectives on biomarker testing in infancy and early autism research^{29–31,63} to develop a comprehensive evidence base of parent acceptability that will inform guidelines for the conduct of early autism research.

OVERALL DESIGN AND STRATEGY

This study utilizes a model of **acceptability** to understand parent perceptions, attitudes, and preferences in the design and conduct of infant ASD identification studies and prediagnostic infant intervention trials. Acceptability is a multi-faceted construct that reflects stakeholders’ anticipated or experiential cognitive/emotional responses to intervention or other clinical encounters. This makes it ideal for a mixed methods approach that integrates multisource data (interviews *and* surveys)^{26,64}. Moreover, robust evidence suggests that acceptability can and *should* be assessed prospectively, prior to engaging in clinical research or servicestively^{26,65}. The **Theoretical Framework of Acceptability (TFA)**²⁶ is an established framework for evaluating prospective and retrospective acceptability. It provides 7 distinct constructs—*affective attitude, burden, ethicality, coherence, opportunity costs, perceived effectiveness, and self-efficacy*—to inform key decisions the conduct and delivery of healthcare interventions. We will use an exploratory sequential mixed methods research design in which qualitative data is collected first to inform the subsequently collected quantitative measure^{66–68}. We use TFA to guide qualitative and quantitative assessments of parent acceptability; items cover the 7 constructs and are organized into 3 operational domains: *perceptions/experiences, attitudes/anticipated reactions, and preferences*.

RECRUITMENT AND FEASIBILITY

A total of 140 parents will be enrolled; recruited from the Carolina Autism and Neurodevelopment (CAN) Research Center at USC (N>2,000; see letter of



support), research studies within the CAN faculty network (N>300), and our own prospective longitudinal studies of ASD (N>150). Given that our ongoing infant longitudinal studies are focused on early identification and biomarker testing (Aim 1), previous or current participation in these studies will be asked and included considered in analyses. We will do the same for any prediagnostic intervention studies or clinical services (but this is expected to be a rare occurrence in this sample). For the qualitative assessment (**N=40**), we will evaluate for saturation at data collection benchmarks (e.g., every 5-10 interviews). If saturation is reached before N=40, we will stop data collection at that point^{69,70}. For the quantitative measure, a distinct, non-overlapping sample of **N=100** (n=50 ASD parents and n=50 NT parents) will be recruited from the same sources. Additionally, samples will be stratified by parents’ *lived experience* (parents with vs without children with ASD) and *current context* (parents with and without infants under 12 months), **Fig 2**. Our sample size for the quantitative assessment analyses is powered on the *Lived Experience* stratification variable (n=50 per group), but we aim to enroll sufficient numbers explore potential moderators that can be followed up on in later studies.

Participant Inclusion/Exclusion Criteria. All participants must be the primary caregiver of at least one child <18 years. ASD parents: at least one child diagnosed with ASD (confirmed via screeners, parent report, and review of diagnostic report when possible^{71,72}). NT parents: no 1st degree relatives with ASD. Reports of non-ASD diagnoses for children (e.g., ADHD) or ASD diagnoses for parents will not be exclusionary, but will be noted for analysis. See *Human Subjects* for full details of inclusion/exclusion criteria. **Racial and Ethnic Representation.** The racial/ethnic characteristics of our current infant sample are more representative compared to other similar studies^{49,61}, as follows: ASD parents: 42% racial/ethnic minorities (24% Black, 10% mixed race, 10% Hispanic); NT parents: 48% racial/ethnic minorities (32% Black, 4% Asian, 8% mixed race, 4% Hispanic). This is representative of our local context in South Carolina. We will make targeted efforts to ensure enrollment that mirrors these demographic proportions. Adequate representation of minority groups will uniquely allow us to examine how parent acceptability intersects with race and other sociodemographic factors. **Recruitment Feasibility.** In addition to recruitment resources mentioned above, we do large recruitment initiatives quarterly, including Simons Powering Autism Research (SPARK) email blasts, South Carolina DDSN direct mailings (N>1200), and social media/Google campaigns. Our local hospital, Prisma Health Data Core, can also provide contact information for all children diagnosed with ASD through the hospital. Our study team maintains a positive, ongoing relationship with participants and our attrition rate is <10% and enrollment rate into supplemental studies for ongoing participants is 98%. Even with multiple stratification variables and our commitment to racial/ethnic representation in our sample, we anticipate we will exceed our enrollment targets. **Operational Feasibility:** The current project capitalizes on the PI’s existing infrastructure and resources (R01MH132925; K23MH120476) that includes computers and audio recorders available for this study. The PI’s lab consists of 3 PhD-level personnel, 1 recruitment coordinator, 4.5 research coordinators, 5 graduate and 20 undergraduate students. For this study, we propose a 6-person study team (3 investigators, 1 consultant, 1 project manager, 1 graduate student) and

ongoing consultation with CAN's Community Advisory Board (CAB) at quarterly meetings and as-needed. **Research Team:** Investigators include a developmental scientist and clinical psychologist with expertise in autism early identification and intervention and infant development (Bradshaw, PI), a public health scientist with expertise in infant clinical trials and mixed methods data collection and analysis (Larson, Co-I), a philosopher and bioethicist with expertise in medical bioethics and translational policy (McClimans, Co-I), and an ASD parent/advocate and geneticist (Beard, Consultant). Discretionary funds and lab/institutional resources allow us to call upon existing, additional staff to assist when needed. Given this significant infrastructure, completion of the project in a two-year period is highly feasible. See *Study Timeline* document for detailed outline of activities.

MEASURES AND DATA ANALYSIS

Qualitative Assessment. The qualitative assessment will be a semi-structured interview with N=40 parents (n=20 ASD, n=20 NT). Parent participants will be the self-identified "primary caregiver" and only 1 per family will be included. Non-parental primary caregivers can be included. In the case of ≥ 2 primary caregivers, a single caregiver will be selected, at random, to participate. We recognize that perspectives of all caregivers are essential for determining acceptability and this will be examined in subsequent, larger studies. Interviewers will use an interview guide that includes primary open-ended questions to address established domains/themes and follow-up probe questions⁷³. Interviews will be conducted in a private location or HIPAA-compliant virtual appointment and last no more than 1 hour. Participants will be compensated \$100 for each interview, regardless of length. Interviews will be audio-recorded (with two devices in case one fails) and an established analytic method for rapid turn-around implementation research, "rapid qualitative analysis," will be used to analyze qualitative data²⁵. **Quantitative Assessment.** To complement the depth of knowledge obtained from qualitative analyses and assess multimethod convergence, quantitative assessments will be collected from N=100 parents. Results of qualitative analysis will guide our selection and adaptation of existing measures of parent perspectives on biomarker testing in infancy and autism early identification research^{29–31,63} that use existing frameworks for ethical research, risk information seeking, and intervention implementation (i.e., the Theoretical Framework of Acceptability (TFA)²⁶, the model for Risk Information Seeking and Processing for preventative health²⁷, and the Consolidated Framework for Implementation Research²⁸). This method will ensure that we are addressing See *Human Subjects* for full details of inclusion/exclusion criteria. constructs that reflect lived experiences of participants. Quantitative assessment items address the 7 component TFA constructs: affective attitude, burden, ethicality, coherence, opportunity costs, perceived effectiveness, and self-efficacy, and are organized into three

TFA Component Construct	Domain	Example Questions 5-point Likert scale: Strongly Disagree to Strongly Agree
AIM 1		
Coherence	Perceptions & Experience	I knew upon enrollment that I might receive clinical research results about my infant indicating a developmental delay and/or ASD.
Affective Attitude	Perceptions & Experience	I had a strong emotional reaction to hearing individual research results about my infant.
Perceived Effectiveness	Perceptions & Experience	Receiving clinical research results as part of this study helped my child.
Perceived Effectiveness	Attitude & Anticipated Reaction	Testing for indicators of ASD at age 3 months could help my child and family.
Ethicality	Preference	I would prefer to wait until after the first birthday for return of any individual results (i.e., feedback about my child's development).
AIM 2		
Perceived Effectiveness	Attitude & Anticipated Reaction	An intervention trial for infants between 3-9 months old showing early signs of ASD could help my child.
Burden	Attitude & Anticipated Reaction	I am concerned about the effect of enrolling into an infant intervention trial might have on my family.
Affective Attitude	Preferences	I would enroll in an infant intervention trial for 3-month-olds if my infant screened positive using a candidate biomarker for ASD, but was not showing any observable developmental delays or ASD behaviors.
Affective Attitude	Preferences	A self-paced, web-based intervention curriculum is my preferred method of receiving support for my 6-month-old infant showing early signs of ASD.

domains: 1) **perceptions and experience**, 2) **attitudes and anticipated reactions**, and 3) **preferences** (see **Table 3**). Identical to our ongoing studies, USC's REDCap system will be used to disseminate the quantitative assessment and store the data. Responses will then be analyzed descriptively and **triangulated** with qualitative results to evaluate convergence⁷⁴. In addition, stratifying variables and demographic characteristics will be incorporated into analyses to further examine individual differences in acceptability. **Sample Size Justification.** N=40 participants for the qualitative assessment is grounded in qualitative methods standards aimed at reaching thematic saturation^{69,70} and is consistent with similar mixed methods studies^{66,75,76}. **Power Analysis** for quantitative data analyses (G*Power parameters: medium

effect size ($d=0.5$; $f^2 = 0.15$), significance $\alpha=.05$, power $\beta=.80$) determined that $N=100$ is sufficient for detecting meaningful differences between subgroups (ANOVAs), associations between variables (linear/logistic regressions), and interactions (multiple regression).

Qualitative Data Analysis. Audio-recorded data from semi-structured interviews will be transcribed using Nvivo (v12) and audited by a team member. Rapid qualitative analysis is appropriate for studies that are shorter-term projects (≤ 1 year), mixed methods designs, and when results inform other aspects of the project^{77,78}, as is the case with the proposed study. The analysis will use a mixed deductive-inductive approach. The deductive component of the analysis will be guided by the TFA constructs (see **Table 3**). Within these domains, themes will be inductively identified. The inductive component will allow themes to emerge organically from the data. In rapid qualitative analysis, each interview is summarized using a template of research-team established and pre-tested domains to reduce the dataset to key information relevant to the research question. Summaries are transferred to respondent-level and/or group level horizontal matrix, by domain. Two independent analysts then review matrices to identify themes. A third analyst guides consensus discussions to determine the final set of themes, which will then be reviewed by all Co-Is and Consultant to ensure that interpretations are validated by a diverse group of experts. Analyses are batched (e.g., groups of 5-10 participants) and the number of themes identified in each round of analysis are tracked to ensure that saturation is reached⁷⁹. Parent acceptability will be explored within and between stratifiers and demographic variables (e.g., ASD vs. NT; racial/ethnic minorities).

Quantitative Data Analysis. Descriptive statistics (means, standard deviations, medians, ranges, frequencies, as appropriate) will be calculated and inferential statistics will be used to test group differences and associations (linear and logistic regressions, chi-squares, ANOVAs, as appropriate), with multiple comparison adjustments as needed. Quantitative assessment responses will be triangulated with qualitative findings and subgroup comparisons will be described to provide insight on parent acceptability of infant ASD identification and prediagnostic intervention research, and real-world implementation.

SPECIFIC AIM OVERVIEW AND OUTCOMES

Acceptability, as defined by TFA domains and constructs (see **Table 3**) will be evaluated with a mixed methods approach. Stratifiers (*lived experience* and *current context* variables) and demographics (race/ethnicity, maternal education, income, # of autistic children) will be considered in all analyses. **Aim 1: Determine parent acceptability of ASD identification research and practice for infants < 12 months.** Parents will report on their **perceptions or experiences**: type of biomarker result received, usefulness of the results for infant and family, next steps taken; **attitude and anticipated reaction** to: receiving positive biomarker result and negative result, receiving false positives/negatives, receiving (or not receiving) predictions about prognosis or future support needs; **preferences** for: infant age biomarker testing/return of results, biomarker type (blood test vs. MRI), acceptable threshold of prediction accuracy, prognostic biomarkers, method of communication about results with researcher/provider, and next action steps. **Aim 2: Evaluate parent acceptability of early-phase development studies and real-world uptake of prediagnostic ASD interventions for infants < 12 months.**

Parents will report on their **perceptions or experiences** of participating in infant intervention trials; **attitude and anticipated reactions** related to: family burden, harmful and beneficial effects intervention on parent-infant relationship, immediately observable gains or lack thereof after intervention onset; **preferences** for: requisite knowledge for informed consent and decision-making, age of intervention onset, intervention method (behavioral vs. medication), intervention targets/content (infant social engagement vs. parent psychoeducation), dosage (duration, frequency), and method of delivery (in-person vs. virtual; parent coaching vs. clinician implementation).

Expected Outcomes. In-depth understanding and a representative evidence base of parent acceptability and associated individual factors are likely to reveal novel insights into the conduct and implementation of infant research that can be examined in follow-up studies with larger samples. Implications of this study extend beyond the field of ASD can guide novel models and hypotheses of parent conceptualization and acceptability of very early identification and intervention for other neurodevelopmental and health conditions. Significantly, this study will lay the empirical groundwork for the development of a comprehensive, participant-driven clinical research toolkit that can be used by investigators to ensure rigorous conduct of ethical research. **Potential Problems & Alternative Strategies.**

This project will be initiated immediately upon funding with existing recruitment and implementation infrastructure. We do not anticipate **recruitment** challenges and this will be monitored weekly to inform timing and scale of larger recruitment initiatives. **Confidentiality.** Some parents who are already enrolled in our ongoing studies may be reticent to describe their experience and attitudes of their research participation to members of our research team. Matching parents with interviewers naïve to their history is feasible and we will ensure that study team interviewers have had no prior interactions with the family in other contexts, and remind families that all data are deidentified and their participation/responses in this study will not in any way impact their participation in, or relationships with, any entities at the university.

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